Vance Vanvolkenburgh
655.662 — Introduction to Healthcare Systems Engineering
Module 6 | M6A2: Shared Vision

Senge defines a shared vision as the answer to one question — what do we want to create? — and insists it is not an idea but a force, drawing its power from common caring. It cannot be issued. It is assembled from the personal visions people already hold, which is why announcing a vision produces compliance while sharing one produces commitment. Chapter 10 ranks the possible responses: commitment, enrollment, genuine compliance, formal compliance, grudging compliance, noncompliance, apathy. Only the first two create anything. The rest keep a program alive without advancing it, and most systems are delivered by people somewhere in the middle of that list.

Stakeholder	Personal vision, in their own terms	What it contributes to the shared vision
Patient	Keep working, keep my evenings, not be a patient all day	The therapy fits the life, not the reverse
Care partner	Not be the one who missed something	The system watches so the family does not have to
Home dialysis nurse	Catch the problem early without drowning in alerts	Attention is a finite clinical resource
Nephrologist	Know whether the prescription change actually worked	Decisions are judged against data, not memory
Home program manager	Grow census without growing technique failure	Retention, not enrollment, is the measure
Dialysis organization	Shift the modality mix and have the economics hold	Home is viable at scale, not just possible
Cycler maker, carrier, EHR vendor	Protect our own roadmaps and release cycles	Interfaces are negotiated, never dictated
Payer (CMS)	Better outcomes for the same dollar	The measure is the patient-year, not the treatment
Engineering team	Build something that is actually used	Adoption is the acceptance criterion

The vision, and why its wording matters. Read down the third column, and one sentence covers all of it: every patient who can dialyze at home does, and stays there. It is intentionally worded positively. The typical default for a monitoring program is a negative vision, do not miss a peritonitis episode, do not allow the completeness metric to drop, and according to Senge (a proponent of positive visions), these negative visions convey a message of helplessness and motivate action only while the danger is apparent. A program operated with “do not let us fail to detect something” generates reluctant compliance from the nurse closing out notifications to free up their queue, which is precisely the kind of behavior that will result in discontinuation of home dialysis.

The four ways in which it supports the overall system. Firstly, it resolves conflicts over requirements which are not resolvable via data. There is no technical solution to the conflict between false notification rate and detection lead time; it has a solution based upon the vision, which says to protect the nurse’s focus, because if she leaves her queue unattended then she is losing the opportunity to have the patient continue with home dialysis as much as if she misses an infection. Secondly, it creates commitment to comply at the point of use, as opposed to

compliance due to nagging. The patient completes the 60-second survey because they wish to continue doing home dialysis, not simply because an application is sending reminders. Thirdly, it is the only means of coordination possible when authority is absent. PD-RPM is an acknowledged system of systems, i.e., neither the manufacturer of the cycler nor the carrier nor the vendor of the EHR may be directed, only coordinated. Finally, when stress arises in engineering design, and in fact stress arises during engineering development as soon as pressure to meet a deadline increases, Senge identifies shared vision as the rudder that should steer the direction of the organization.

An example: The week that the threshold for the alert was negotiated. During the integration readiness meeting, analytics recommended reducing the alert threshold for ultrafiltration decline from 20 percent to 12 percent, since it would identify more declining patients. However, the nurse lead protested: At 150 patients and one full-time equivalent nurse, it would approximately triple the amount of queued activity. Without a shared vision this represents a standoff resolved by rank or by volume. When one party loses they will comply formally but cease participating. Since both parties were now advocating positions about the same outcome, the issue became answerable with respect to a shared vision, namely which threshold would keep more patients at home after 12 months? Neither side had those numbers, thus the review asked for them rather than making a decision. In turn, the interim threshold remained at 20 percent with a trend direction qualifier, proposed by the nurse lead who had come to oppose and departed having created the compromise. This is the transition from genuine compliance to enrollment and it generated a better standard than either initial proposal.

The distance between the vision and the roughly one in eight patients who are currently engaged in home dialysis is large enough to be discouraging; this is Senge’s most frequent reason for a vision to expire. When held directly against current performance it represents creative tension, an energy-generating element rather than a justification to lower goals. This is why shared vision is relevant to this module. Advanced development is exactly where conflicting individual visions collide regarding what to create, and a program without anything to resolve these disputes will resolve them by seniority.